THORACIC TRAUMA — Clinical Revision Table

Bold = high-yield. 2nd leading cause trauma death; 25% of trauma deaths from chest injury. Early deaths often PREVENTABLE (tension pneumo, tamponade, airway, haemorrhage).

CHEST WALL

Injury	Key features / signs	Management	Pearls
Rib fractures	Heal 3–6 wks. Ribs 4–9 commonest. **Lower ribs (X–XII)** → **liver/spleen/kidney**. **1st/2nd # + apical cap** → **great vessel injury**	Pain control, pulmonary function. Repeat CXR 3h (delayed haemo/pneumo). **Elderly + multiple #** → **admit (pneumonia)**	Splinting → atelectasis/pneumonia
Flail chest	**≥3 ribs in ≥2 places** → **free segment** → **PARADOXICAL respiration** (in on inspiration). Associated pulmonary contusion (real problem)	Pain control, pulmonary toilet, O2, treat contusion. Severe → intubation + IPPV (internal pneumatic fixation)	Contusion + hypoxaemia kill, not the flail
Sternal fracture	Anterior blunt / **seatbelt deceleration**. Low mortality. → myocardial contusion risk	Lateral X-ray; **screen myocardial contusion (ECG + serial cardiac enzymes), monitor 24–48h**; analgesia	NO association with blunt aortic injury
Subcutaneous emphysema	Air in chest wall → marker of serious injury	Benign + self-limiting → **high-flow O2**. Massive → 'gills' incisions	—

PLEURAL / LUNG

Injury	Key features / signs	Management	Pearls
Pulmonary contusion	Alveolar oedema/haemorrhage; hypoxaemia ~50%; **X-ray findings within minutes, ARDS-like**	**Restrict IV fluids**, pulmonary toilet, pain control, respiratory support. **NO prophylactic antibiotics**. Mortality 5–16%	Worsens over 24–48h
Simple pneumothorax	Air in pleural space; hyperresonance, ↓breath sounds. Grades: small/large/occult	Small/occult → observe. **Moderate/large** → **chest tube (5th ICS, anterior axillary, safe triangle)**	—
Open pneumothorax	**'Sucking chest wound'**; chest wall defect; respiratory distress	**Three-way occlusive dressing** (prevents tension) → chest tube → definitive closure	Penetrating > blunt
Tension pneumothorax	EMERGENCY. One-way valve → mediastinal shift. **Distended neck veins, hypotension, tracheal deviation (contralateral), absent breath sounds, hyperresonance**	**CLINICAL dx (before CXR). Immediate needle decompression: 2nd ICS, mid-clavicular line, 16G** → **then chest tube**	↓venous return → shock
Haemothorax	Blood in pleura (lung = commonest source, self-limiting). ↓breath sounds + **DULLNESS**. X-ray >200mL	**Chest tube. Thoracotomy if: initial >1500mL OR >200mL/h ×2–3h OR persistent transfusion**	Massive = up to 3L

AIRWAY / OESOPHAGUS / DIAPHRAGM

Injury	Key features / signs	Management	Pearls
Tracheobronchial injury	MVC; **80% within 2.5cm of carina**; massive air leak, haemoptysis, sub-cut emphysema. **Persistent large pneumo + air leak after chest tube**	**Bronchoscopy = definitive dx**. Intubation beyond injury; surgical repair (thoracotomy)	Mortality 10%
Oesophageal perforation	Penetrating; high morbidity, mediastinitis. **Gastric content in chest tube**, pneumomediastinum, effusion without chest injury	Surgical repair; antibiotics	2nd to penetrating injury
Diaphragmatic tear	Severe trauma + associated injuries	**Mandates abdominal exploration**	—

CARDIOVASCULAR

Injury	Key features / signs	Management	Pearls
Myocardial contusion	>55% of severe closed chest trauma	Monitor; **ECG + serial troponin** (normal at 4–6h excludes); O2, analgesia	Risk stratify with troponin
Penetrating cardiac / tamponade	**RV most common (43%; LV 34%)**; suspect in 'the BOX' (clavicles–mid-clavicular–costal margins). **Beck's triad: muffled heart sounds + distended neck veins + hypotension**; narrow pulse pressure, pulsus paradoxus	**ECHO** (effusion + RV collapse); ECG electrical alternans. Fluids; **pericardiocentesis** (blunt/stable); **thoracotomy + repair** (penetrating)	80% die at scene
Blunt aortic injury	**Most common vessel injured by blunt trauma. 80–90% at descending aorta distal to left subclavian (isthmus)**. CXR: **widened mediastinum >8cm**, loss of aortic knob, apical cap, L effusion	**CT angiography (helical) = dx.** Fix ABCDE first; **control BP (SBP 100–120, β-blocker/esmolol)**; surgery or endovascular repair	85% survive if prompt; paraplegia 5–7%

EMERGENCY DEPARTMENT THORACOTOMY

Indications	Cardiac arrest in **PENETRATING** injury, massive haemothorax, penetrating + tamponade, large open wound, major vascular/tracheobronchial injury, oesophageal perforation. **CONTRAINDICATED in pulseless BLUNT trauma**

5 moves	1. Incise inferior pulmonary ligament · 2. Open pericardium + staple/suture cardiac laceration · 3. Open cardiac massage · 4. Clamp pulmonary hilum / twist bleeding lung · 5. **Clamp thoracic aorta**